

WEF

Administrator: Windows PowerShell (x86)

```
PS C:\Users\Administrator> hostname
WEF-Marcin
PS C:\Users\Administrator> ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : eng-dhcp.opnsense
    Link-local IPv6 Address . . . . . : fe80::9b2c:76f7:af8e:ee6a%12
    IPv4 Address. . . . . : 192.168.222.144
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.222.247
PS C:\Users\Administrator> type C:\Windows\System32\drivers\etc\hosts
# Copyright (c) 1993-2009 Microsoft Corp.

# This is a sample HOSTS file used by Microsoft TCP/IP for Windows.

# This file contains the mappings of IP addresses to host names. Each
# entry should be kept on an individual line. The IP address should
# be placed in the first column followed by the corresponding host name.
# The IP address and the host name should be separated by at least one
# space.

# Additionally, comments (such as these) may be inserted on individual
# lines or following the machine name denoted by a '#' symbol.

# For example:
#
#     102.54.94.97     rhino.acme.com   # source server
#     38.25.63.10     x.acme.com      # x client host

# localhost name resolution is handled within DNS itself.
#
#     127.0.0.1       localhost
#     ::1             localhost
192.168.222.144      WEF-Marcin
192.168.222.142      WEC-Marcin
PS C:\Users\Administrator>
```

WEC

```
Administrator: Windows PowerShell (x86)
PS C:\Users\Administrator> hostname
WEC-Marcin
PS C:\Users\Administrator> ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : eng-dhcp.opnsense
    Link-local IPv6 Address . . . . . : fe80::e880:fe46:6e4b:c8f3%12
    IPv4 Address. . . . . : 192.168.222.142
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.222.247
PS C:\Users\Administrator> type C:\Windows\System32\drivers\etc\hosts
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#
# This is a sample HOSTS file used by Microsoft TCP/IP for Windows.
#
# This file contains the mappings of IP addresses to host names. Each
# entry should be kept on an individual line. The IP address should
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# space.
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# lines or following the machine name denoted by a '#' symbol.
#
# For example:
#
#       102.54.94.97     rhino.acme.com   # source server
#       38.25.63.10     x.acme.com      # x client host
#
# localhost name resolution is handled within DNS itself.
#       127.0.0.1       localhost
#       ::1             localhost
192.168.222.144        WEF-Marcin
192.168.222.142        WEC-Marcin
PS C:\Users\Administrator>
```

```
New-NetFirewallRule -DisplayName "WEF WinRM HTTPS 5986" -Direction
Inbound -Protocol TCP -LocalPort 5986 -Action Allow -Profile Any
```

Verify:

```
Get-NetFirewallRule -DisplayName "WEF WinRM HTTPS 5986"
```

Create Folder

```
New-Item -ItemType Directory -Path C:\Certs -Force | Out-Null
```

Root CA

```
$rootCA = New-SelfSignedCertificate `
    -Type Custom -Subject "CN=Marcin-Lab-RootCA" `
    -KeyUsage CertSign,CRLSign,DigitalSignature `
    -KeyExportPolicy Exportable -KeyLength 2048 -HashAlgorithm SHA256 `
    -NotAfter (Get-Date).AddYears(10) `
```

```
-CertStoreLocation Cert:\LocalMachine\My `
-TextExtension @("2.5.29.19={text}CA=1")
```

Server cert for WEC-Marcin

```
$serverCert = New-SelfSignedCertificate `
  -Type Custom -Subject "CN=WEC-Marcin" -DnsName "WEC-
Marcin","192.168.222.142" `
  -KeyExportPolicy Exportable -KeyLength 2048 -HashAlgorithm SHA256 `
  -NotAfter (Get-Date).AddYears(5) `
  -CertStoreLocation Cert:\LocalMachine\My -Signer $rootCA `
  -TextExtension @("2.5.29.37={text}1.3.6.1.5.5.7.3.1")
```

Client cert for WEF-Marcin

```
$clientCert = New-SelfSignedCertificate `
  -Type Custom -Subject "CN=WEF-Marcin" `
  -KeyExportPolicy Exportable -KeyLength 2048 -HashAlgorithm SHA256 `
  -NotAfter (Get-Date).AddYears(5) `
  -CertStoreLocation Cert:\LocalMachine\My -Signer $rootCA `
  -TextExtension @("2.5.29.37={text}1.3.6.1.5.5.7.3.2")
```

Export Root CA public cert

```
Export-Certificate -Cert $rootCA -FilePath C:\Certs\RootCA.cer | Out-Null
```

Export client cert with private key for transfer to WEF-Marcin

```
$pwd = ConvertTo-SecureString -String "TempP@ssw0rd!" -Force -AsPlainText
Export-PfxCertificate -Cert $clientCert -FilePath C:\Certs\WEF-Marcin-
client.pfx -Password $pwd | Out-Null
```

Import Root CA into this machine's own Trusted Root

```
Import-Certificate -FilePath C:\Certs\RootCA.cer -CertStoreLocation
Cert:\LocalMachine\Root | Out-Null
```

```
Write-Host "`n=== SAVE THESE THUMBPRINTS ===" -ForegroundColor Yellow
Write-Host "Root CA:      $($rootCA.Thumbprint)"
Write-Host "Server cert: $($serverCert.Thumbprint)"
Write-Host "Client cert: $($clientCert.Thumbprint)"
```

```
Root CA:      B551F8F7DEEA761FF7D1958DB4B835F1AD043F04
Server cert:  B9844DEF377382027F2F646516803BEF714F22EE
Client cert:  EBCF3D97349DE1E321A32056C3DC65C8218D15C6
```

Step 3 — Finish setup on WEC-Marcin (same machine, same session)

Rebind the HTTPS listener to the new server cert:

powershell

```
winrm delete winrm/config/Listener?Address=*&Transport=HTTPS 2>$null

New-Item -Path WSMan:\localhost\Listener -Transport HTTPS -Address * `
-CertificateThumbPrint "B9844DEF377382027F2F646516803BEF714F22EE" -Force
```

Re-apply the negotiate-client-cert fix at the HTTP.SYS level (this was the earlier fix — needs to be redone since we deleted the listener):

powershell

```
netsh http show sslcert ipport=0.0.0.0:5986
```

Note the Application ID GUID shown, then:

powershell

```
netsh http delete sslcert ipport=0.0.0.0:5986

netsh http add sslcert ipport=0.0.0.0:5986 `
  certhash=B9844DEF377382027F2F646516803BEF714F22EE `
  appid="{PASTE_APPID_GUID_HERE}" `
  clientcertnegotiation=enable
```

```
SSL Certificate bindings:
-----

IP:port           : 0.0.0.0:5986
Certificate Hash   : b9844def377382027f2f646516803bef714f22ee
Application ID     : {afebb9ad-9b97-4a91-9ab5-daf4d59122f6}
Certificate Store Name : (null)
Verify Client Certificate Revocation : Enabled
Verify Revocation Using Cached Client Certificate Only : Disabled
Usage Check       : Enabled
Revocation Freshness Time : 0
URL Retrieval Timeout : 0
Ctl Identifier     : (null)
Ctl Store Name     : (null)

Administrator: Windows PowerShell (x86)

PS C:\Users\Administrator> netsh http delete sslcert ipport=0.0.0.0:5986

SSL Certificate successfully deleted

PS C:\Users\Administrator> netsh http add sslcert ipport=0.0.0.0:5986 `
>>   certhash=B9844DEF377382027F2F646516803BEF714F22EE `
>>  appid="{afebb9ad-9b97-4a91-9ab5-daf4d59122f6}" `
>>   clientcertnegotiation=enable

SSL Certificate successfully added
```

Ensure service-level cert auth is still on:

powershell

```
Set-Item WSMan:\localhost\Service\Auth\Certificate -Value $true
```

Recreate the local account + certmapping with the new thumbprints:

powershell

```
# Skip if WEFClient user already exists and is fine
$pw = ConvertTo-SecureString "AnotherStr0ngP@ss!" -AsPlainText -Force
New-LocalUser -Name "WEFClient" -Password $pw -PasswordNeverExpires -
UserMayNotChangePassword -ErrorAction SilentlyContinue
Add-LocalGroupMember -Group "Remote Management Users" -Member "WEFClient" -
ErrorAction SilentlyContinue

winrm create "winrm/config/service/certmapping?
Issuer=B551F8F7DEEA761FF7D1958DB4B835F1AD043F04+Subject=CN=WEF-Marcin+URI=*"
"@{Username=`"WEFClient`";Password=`"AnotherStr0ngP@ss!`"}`"

winrm enumerate winrm/config/service/certmapping
```

If you cannot create user and add user to local group

```

lyContinue ^C
PS C:\Users\Administrator> ^C
PS C:\Users\Administrator> net user WEFClient AnotherStr0ngP@ss! /ad
The password entered is longer than 14 characters. Computers
with Windows prior to Windows 2000 will not be able to use
this account. Do you want to continue this operation? (Y/N) [Y]: Y
The command completed successfully.

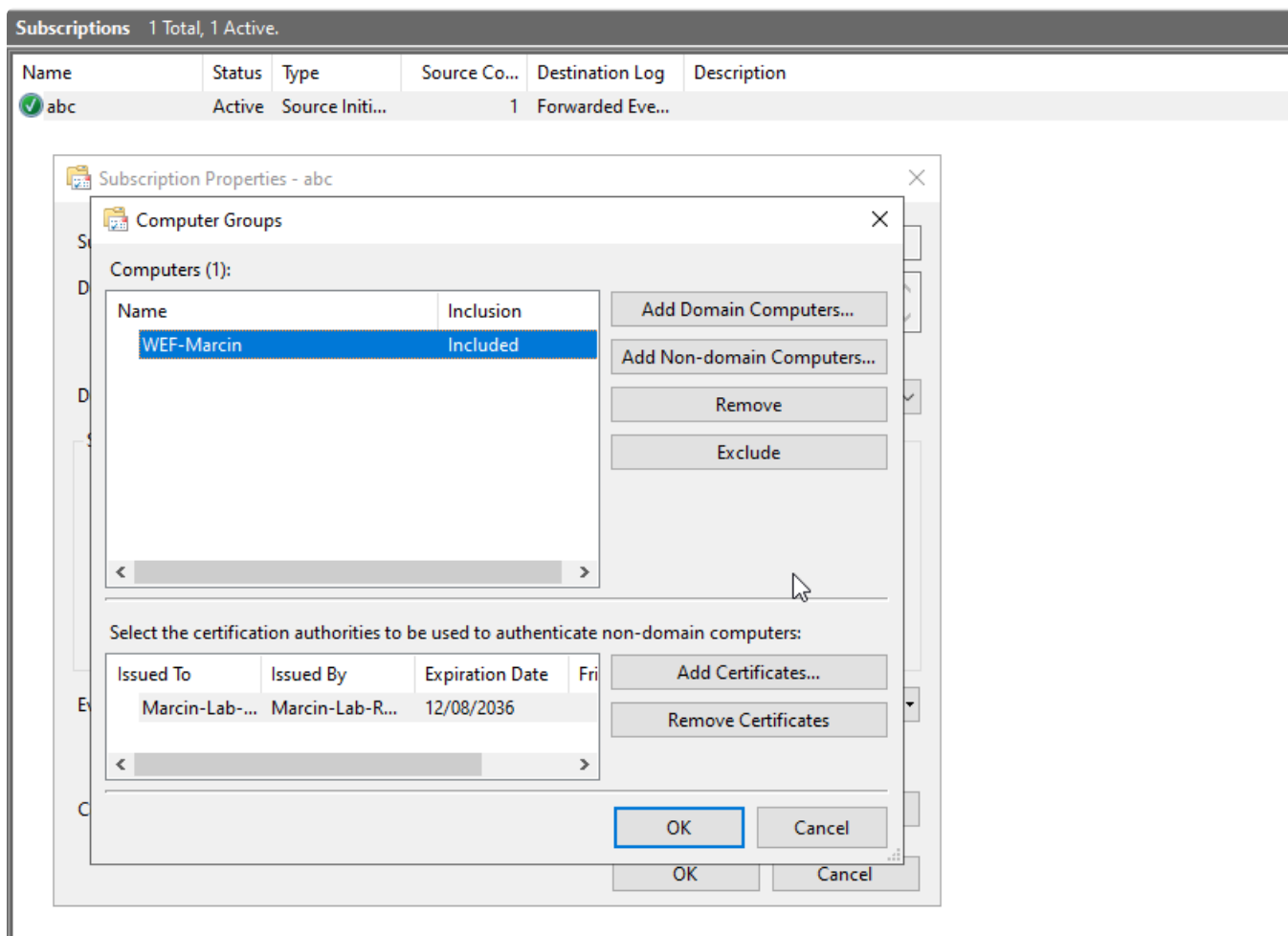
```

```

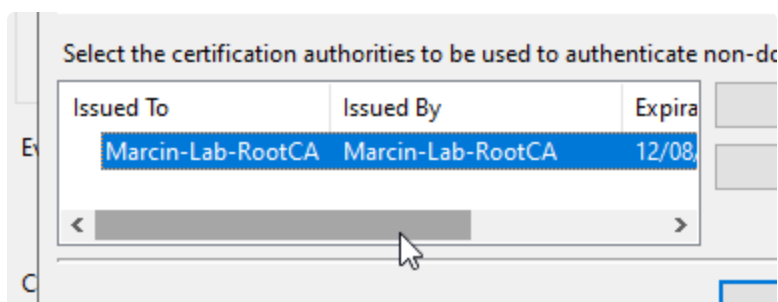
PS C:\Users\Administrator> winrm enumerate winrm/config/service/certmapping
PS C:\Users\Administrator> net localgroup "Remote Management Users" WEFClient /add
The command completed successfully.

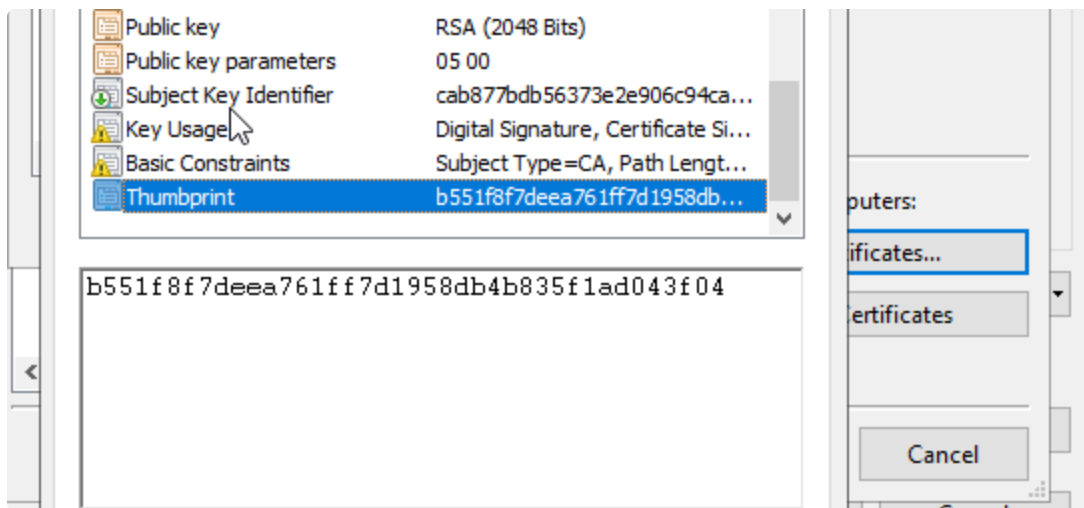
```

Create Subscription



Find correct Certs





Step 4 — Set up WEF-Marcin

Copy `C:\Certs\RootCA.cer` and `C:\Certs\WEF-Marcin-client.pfx` to WEF-Marcin now, then run:

powershell

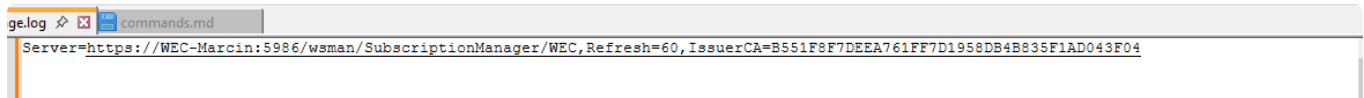
```
Import-Certificate -FilePath C:\Certs\RootCA.cer -CertStoreLocation
Cert:\LocalMachine\Root

$pwd = ConvertTo-SecureString -String "TempP@ssw0rd!" -Force -AsPlainText
Import-PfxCertificate -FilePath C:\Certs\WEF-Marcin-client.pfx -
CertStoreLocation Cert:\LocalMachine\My -Password $pwd

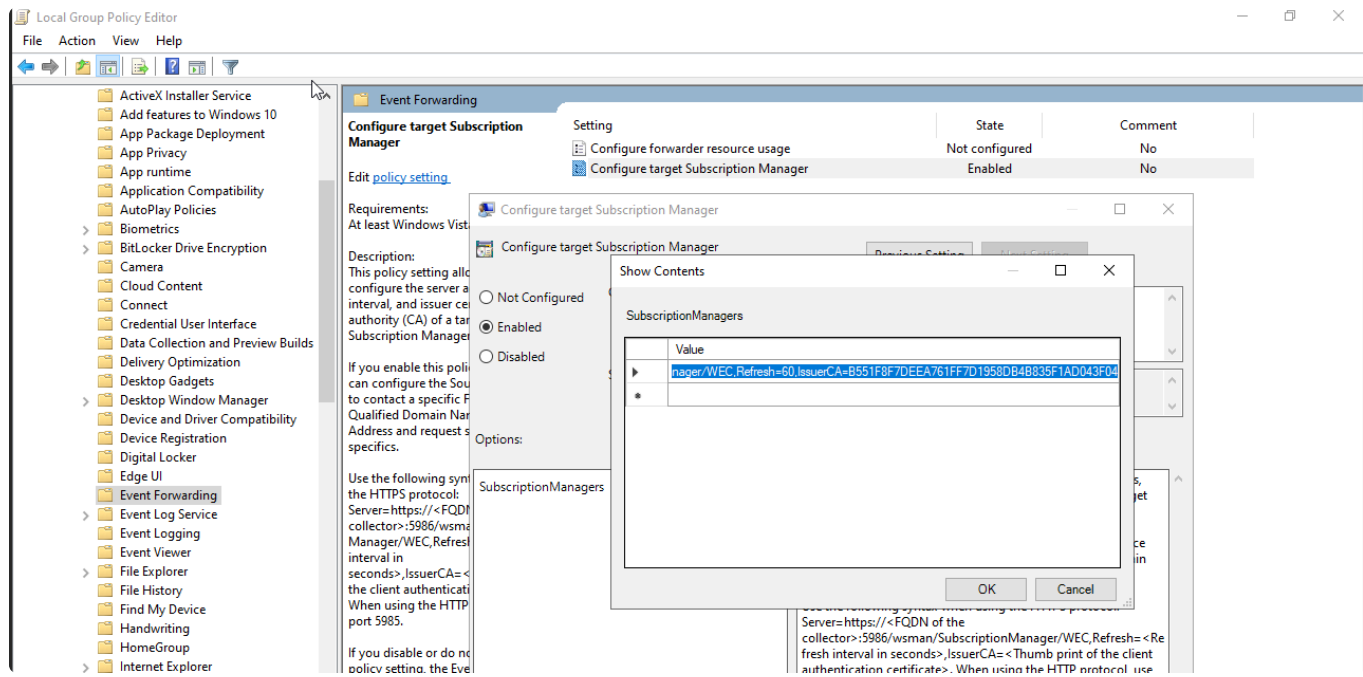
Get-ChildItem Cert:\LocalMachine\My | Where-Object {$_.Subject -eq "CN=WEF-
Marcin"} | Select Subject, Thumbprint, HasPrivateKey
```

Confirm the thumbprint shown is `EBCF3D97349DE1E321A32056C3DC65C8218D15C6` with `HasPrivateKey: True`.

Update the SubscriptionManager (via gpedit, same dialog as before, or registry directly):



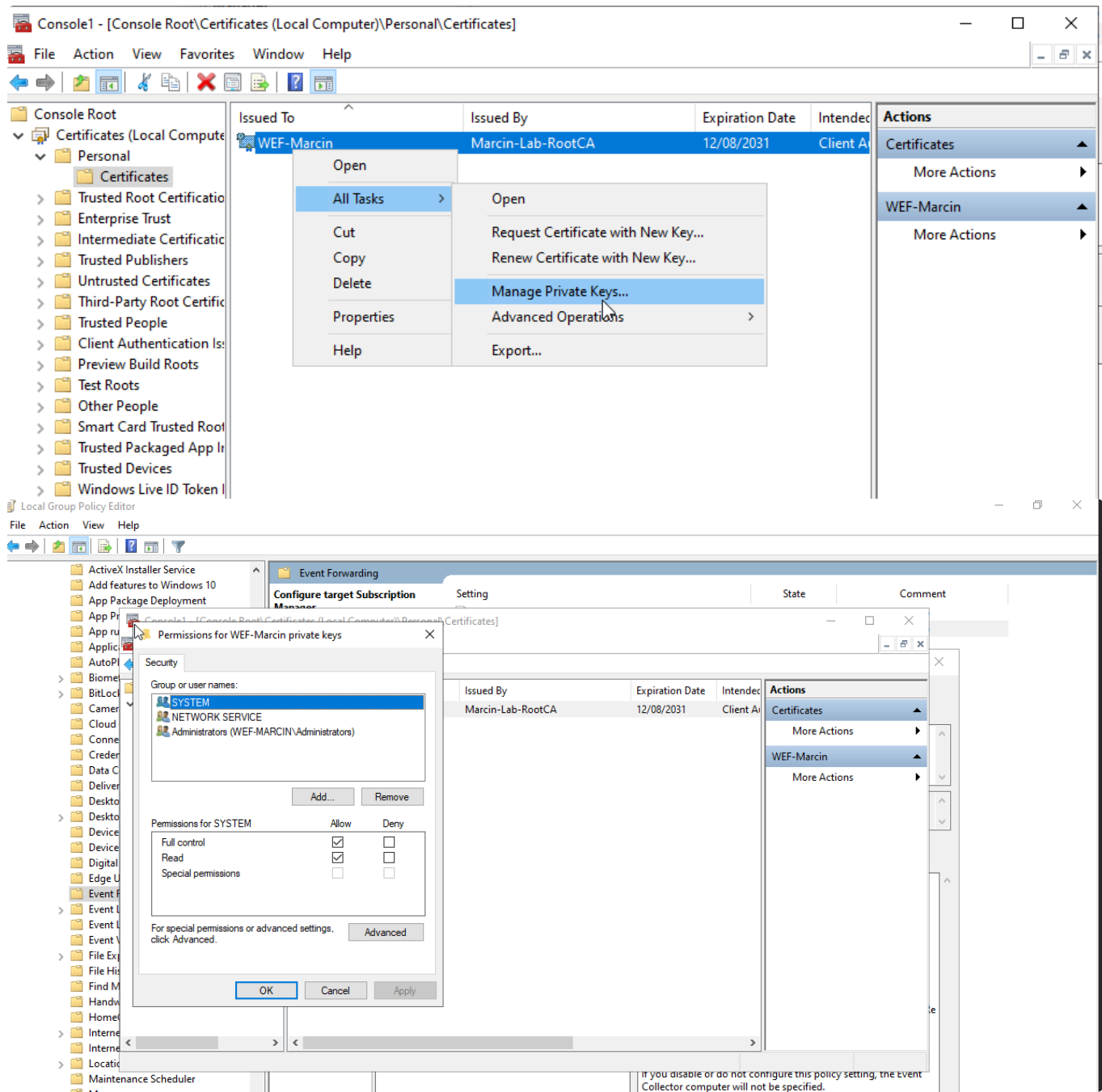
Enable Configure target Subscription Manager



gpupdate /force

Restart-Service EventLog -Force

Go to mmc add Read permission to read key for NETWORK-SERVICE



Test
WEF

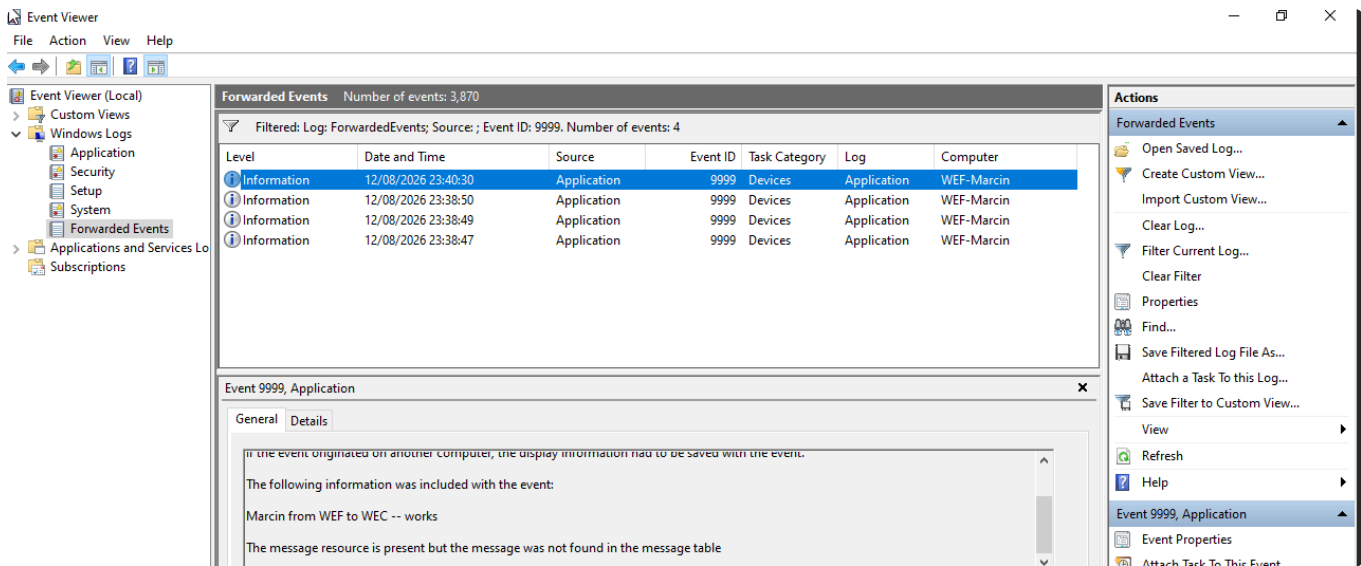
```
Administrator: Windows PowerShell (x86)
PS C:\Users\Administrator> Write-EventLog -LogName Application -Source "Application" -EventId 9999 -EntryType Information -Message "Marcin from WEF to WEC -- works"
PS C:\Users\Administrator> Write-EventLog -LogName Application -Source "Application" -EventId 9999 -EntryType Information -Message "Marcin from WEF to WEC -- works"
PS C:\Users\Administrator> Write-EventLog -LogName Application -Source "Application" -EventId 9999 -EntryType Information -Message "Marcin from WEF to WEC -- works"
PS C:\Users\Administrator> Write-EventLog -LogName Application -Source "Application" -EventId 9999 -EntryType Information -Message "Marcin from WEF to WEC -- works"
PS C:\Users\Administrator> hostname
WEF-Marcin
PS C:\Users\Administrator> ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : eng-dhcp.opnsense
    Link-local IPv6 Address . . . . . : fe80::9b2c:76f7:af8e:ee6a%12
    IPv4 Address. . . . . : 192.168.222.144
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.222.247
PS C:\Users\Administrator>
```

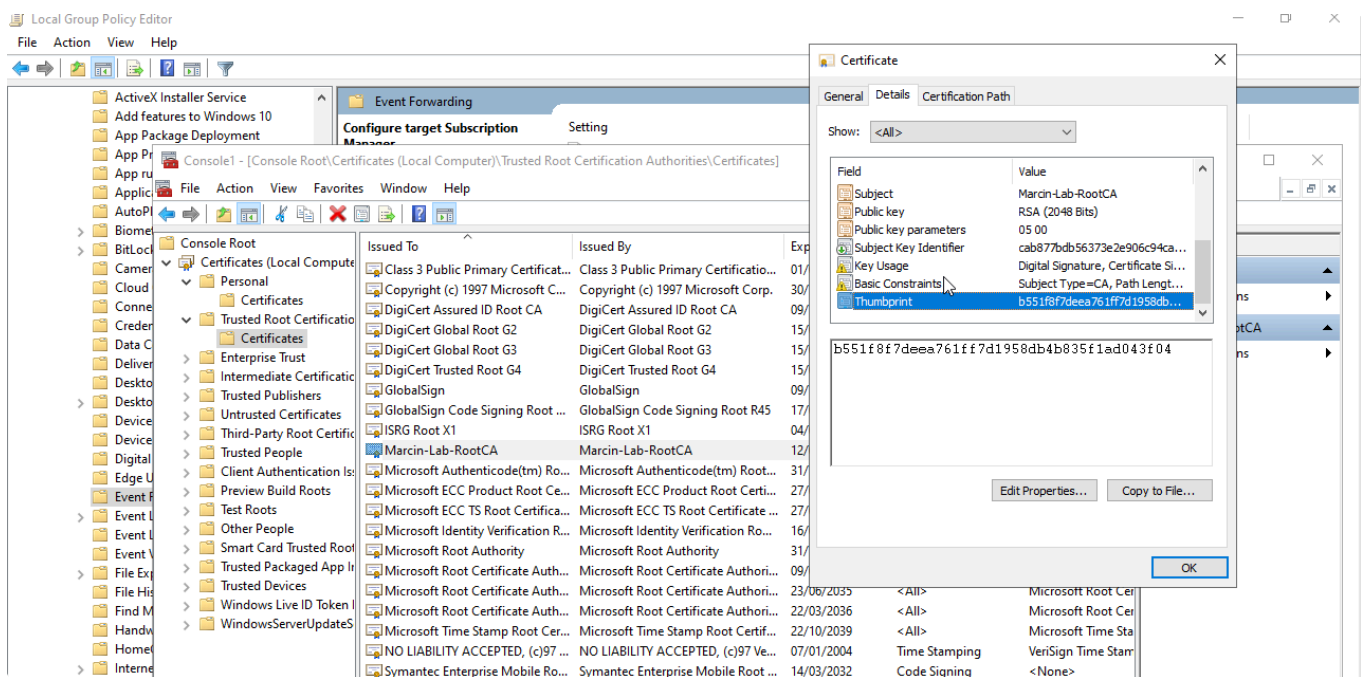
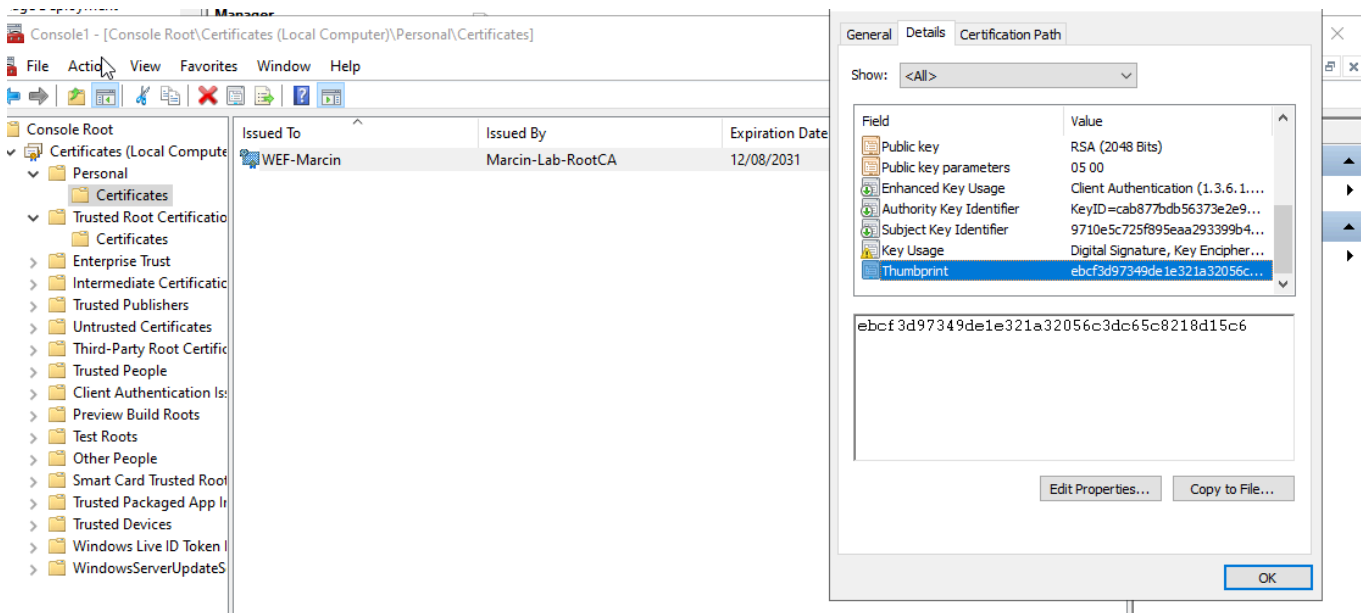
WEC



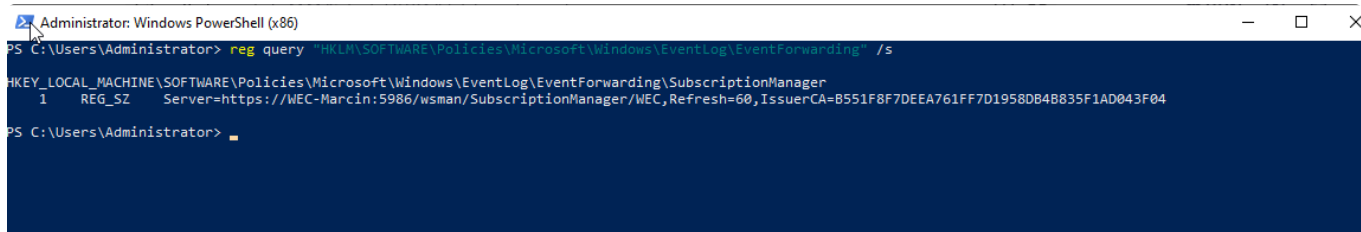
Validate settings

```
Root CA:      B551F8F7DEEA761FF7D1958DB4B835F1AD043F04
Server cert:  B9844DEF377382027F2F646516803BEF714F22EE
Client cert:  EBCF3D97349DE1E321A32056C3DC65C8218D15C6
```

WEF



```
reg query "HKLM\SOFTWARE\Policies\Microsoft\Windows\EventLog\EventForwarding" /s
```



```
Administrator: Windows PowerShell (x86)
PS C:\Users\Administrator> net localgroup "Event Log Readers"
Alias name      Event Log Readers
Comment        Members of this group can read event logs from local machine

Members

-----
NT AUTHORITY\NETWORK SERVICE
The command completed successfully.

PS C:\Users\Administrator>
```

WEC

Permission to Read Event Log

```
Administrator: Windows PowerShell (x86)
PS C:\Users\Administrator> net localgroup "Event Log Readers"
Alias name      Event Log Readers
Comment        Members of this group can read event logs from local machine

Members

-----
NT AUTHORITY\NETWORK SERVICE
The command completed successfully.

PS C:\Users\Administrator>
```

Subscription

```
C:\> Administrator: Command Prompt

C:\share-Marcin> wecutil es
abc

C:\share-Marcin> wecutil gs abc > abc.xml

C:\share-Marcin> type abc.xml
Subscription Id: abc
SubscriptionType: SourceInitiated
Description:
Enabled: true
Uri: http://schemas.microsoft.com/wbem/wsman/1/windows/EventLog
ConfigurationMode: Normal
DeliveryMode: Push
DeliveryMaxLatencyTime: 900000
HeartbeatInterval: 900000
Query: <QueryList><Query Id="0"><Select Path="Application">*</Select><Select Path="Security">*</Select><Select Path="ForwardedEvents">*</Select><Select Path="HardwareEvents">*</Select><Select Path="Microsoft-Windows-AppV-Client/Admin">*</Select><Select Path="Microsoft-Windows-AppV-Client/Virtual Applications">*</Select><Select Path="Microsoft-Windows-ServerCore-ShellLaunchDiagnosticInvoker/Operational">*</Select><Select Path="Microsoft-Windows-Experience-Virtualization
```

```

ect><Select Path="Microsoft-Windows-WinINet-Capture/Analytic">*</Select><Select Path="Microsoft-Windows-WinINet/Operational">*</Select><Select Path="Microsoft-Windows-Winlogon/Operational">*</Select><Select Path="Microsoft-Windows-WinNat/Operational">*</Select><Select Path="Microsoft-Windows-Winsock-WS2HELP/Operational">*</Select><Select Path="Microsoft-Windows-Winsock-NameResolution/Operational">*</Select><Select Path="Microsoft-Windows-Winsock-AFD/Operational">*</Select><Select Path="Microsoft-Windows-Wired-AutoConfig/Operational">*</Select><Select Path="Microsoft-Windows-WMI-Activity/Operational">*</Select><Select Path="Microsoft-Windows-WPNSS-Service/Operational">*</Select><Select Path="Microsoft-Windows-Wordpad/Admin">*</Select><Select Path="Microsoft-Windows-Workplace Join/Admin">*</Select><Select Path="Microsoft-Windows-WPD-ClassInstaller/Operational">*</Select><Select Path="Microsoft-Windows-WPD-CompositeClassDriver/Operational">*</Select><Select Path="Microsoft-Windows-WPD-MTPClassDriver/Operational">*</Select><Select Path="OpenSSH/Admin">*</Select><Select Path="OpenSSH/Operational">*</Select><Select Path="Windows PowerShell">*</Select></Query></QueryList>ReadExistingEvents: false
TransportName: HTTPS
ContentFormat: RenderedText
Locale: en-IE
LogFile: ForwardedEvents
PublisherName: Microsoft-Windows-EventCollector
AllowedIssuerCAList:
    B551F8F7DEEA761FF7D1958D84B835F1AD043F04
AllowedSubjectList:
    WEF-Marcin
DeniedSubjectList:
AllowedSourceDomainComputers:

EventSource[0]:
    Address: WEF-Marcin
    Enabled: true

C:\share-Marcin>

```

ssllcert

```

Administrator: Command Prompt

C:\share-Marcin>hostname
WEF-Marcin

C:\share-Marcin>netsh http show sslcert ipport=0.0.0.0:5986

SSL Certificate bindings:
-----

IP:port                : 0.0.0.0:5986
Certificate Hash       : b9844def377382027f2f646516803bef714f22ee
Application ID        : {afebb9ad-9b97-4a91-9ab5-daf4d59122f6}
Certificate Store Name : (null)
Verify Client Certificate Revocation : Enabled
Verify Revocation Using Cached Client Certificate Only : Disabled
Usage Check           : Enabled
Revocation Freshness Time : 0
URL Retrieval Timeout : 0
Ctl Identifier        : (null)
Ctl Store Name       : (null)
DS Mapper Usage       : Disabled
Negotiate Client Certificate : Enabled
Reject Connections    : Disabled
Disable HTTP2         : Not Set
Disable QUIC          : Not Set
Disable TLS1.2        : Not Set
Disable TLS1.3        : Not Set
Disable OCSP Stapling : Not Set
Enable Token Binding  : Not Set
Log Extended Events   : Not Set
Disable Legacy TLS Versions : Not Set
Enable Session Ticket : Not Set
Enable Caching Client Hello : Not Set
Extended Properties:

```

winrm

```

C:\share-Marcin>hostname
WEC-Marcin

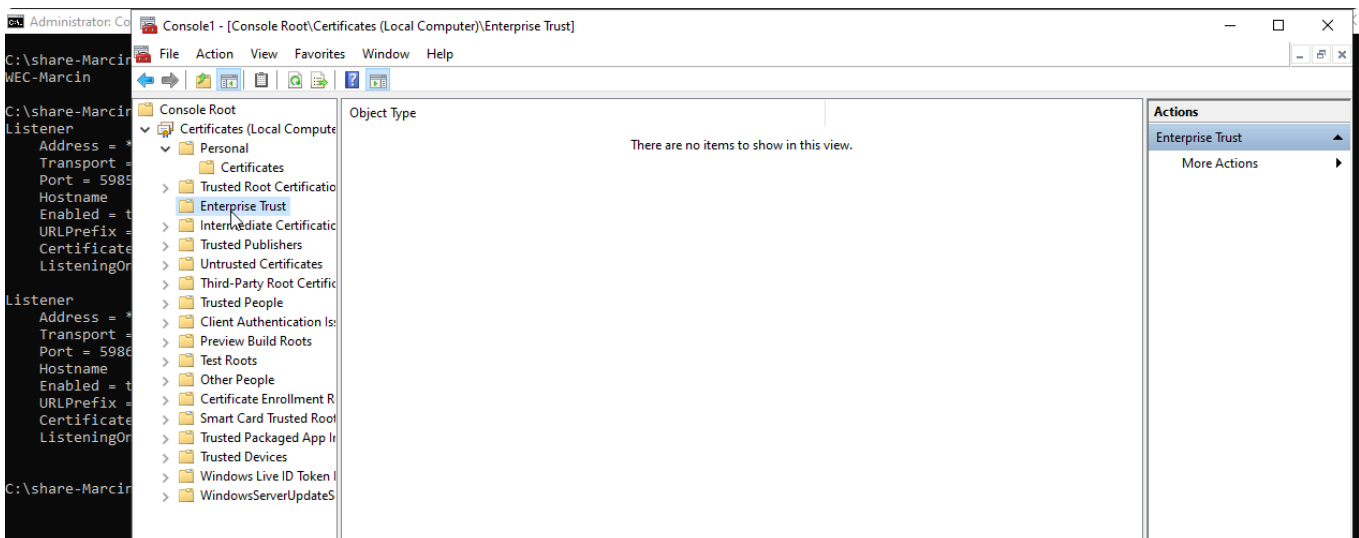
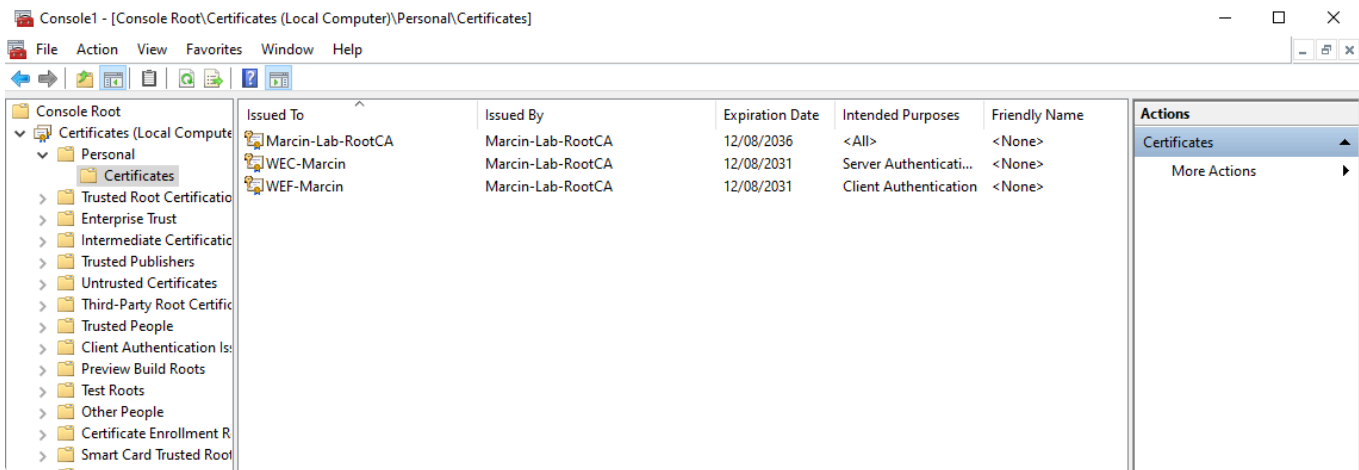
C:\share-Marcin>winrm e winrm/config/listener
Listener
  Address = *
  Transport = HTTP
  Port = 5985
  Hostname
  Enabled = true
  URLPrefix = wsman
  CertificateThumbprint
  ListeningOn = 127.0.0.1, 192.168.222.142, ::1, fe80::e880:fe46:6e4b:c8f3%12

Listener
  Address = *
  Transport = HTTPS
  Port = 5986
  Hostname
  Enabled = true
  URLPrefix = wsman
  CertificateThumbprint = B9844DEF377382027F2F646516803BEF714F22EE
  ListeningOn = 127.0.0.1, 192.168.222.142, ::1, fe80::e880:fe46:6e4b:c8f3%12

C:\share-Marcin>

```

Certs



Administrator: Co... Console1 - [Console Root\Certificates (Local Computer)\Personal\Certificates]

File Action View Favorites Window Help

C:\share-Marcin
WEC-Marcin

C:\share-Marcin
Listener
Address = *
Transport =
Port = 5985
Hostname
Enabled = t
URLPrefix =
Certificate
ListeningOn

Console Root
Certificates (Local Computer)
Personal
Certificates
Trusted Root Certification Authorities
Enterprise Trust
Intermediate Certification Authorities
Trusted Publishers
Untrusted Certificates
Third-Party Root Certificates
Trusted People
Client Authentication Issuance Policies
Preview Build Roots
Test Roots
Other People
Certificate Enrollment Rights
Smart Card Trusted Root Certificates
Trusted Packaged Applications
Trusted Devices
Windows Live ID Token Issuance Policies
WindowsServerUpdateServices

Issued To	Issued By	Expiration Date	Intended Purposes	Friendly Name
Marcin-Lab-RootCA	Marcin-Lab-RootCA	12/08/2036	<All>	<None>
WEC-Marcin	Marcin-Lab-RootCA	12/08/2031	Server Authentication	<None>
WEC-Marcin	Marcin-Lab-RootCA	12/08/2031	Client Authentication	<None>

Certificate

General Details Certification Path

Show: <All>

Field	Value
Subject	Marcin-Lab-RootCA
Public key	RSA (2048 Bits)
Public key parameters	05 00
Subject Key Identifier	cab877bdb56373e2e906c94ca...
Key Usage	Digital Signature, Certificate Si...
Basic Constraints	Subject Type=CA, Path Leng...
Thumbprint	b551f8f7deea761ff7d1958db4b835f1ad043f04

b551f8f7deea761ff7d1958db4b835f1ad043f04

Edit Properties Convert to File

Actions
Certificates
More Actions
Marcin-Lab-RootCA
More Actions

Administrator: Co... Console1 - [Console Root\Certificates (Local Computer)\Personal\Certificates]

File Action View Favorites Window Help

C:\share-Marcin
WEC-Marcin

C:\share-Marcin
Listener
Address = *
Transport =
Port = 5985
Hostname
Enabled = t
URLPrefix =
Certificate
ListeningOn

Console Root
Certificates (Local Computer)
Personal
Certificates
Trusted Root Certification Authorities
Enterprise Trust
Intermediate Certification Authorities
Trusted Publishers
Untrusted Certificates
Third-Party Root Certificates
Trusted People
Client Authentication Issuance Policies
Preview Build Roots
Test Roots
Other People
Certificate Enrollment Rights
Smart Card Trusted Root Certificates
Trusted Packaged Applications
Trusted Devices
Windows Live ID Token Issuance Policies
WindowsServerUpdateServices

Issued To	Issued By	Expiration Date	Intended Purposes	Friendly Name
Marcin-Lab-RootCA	Marcin-Lab-RootCA	12/08/2036	<All>	<None>
WEC-Marcin	Marcin-Lab-RootCA	12/08/2031	Server Authentication	<None>
WEC-Marcin	Marcin-Lab-RootCA	12/08/2031	Client Authentication	<None>

Certificate

General Details Certification Path

Show: <All>

Field	Value
Public key parameters	05 00
Subject Alternative Name	DNS Name=WEC-Marcin, DNS ...
Enhanced Key Usage	Server Authentication (1.3.6...
Authority Key Identifier	KeyID=cab877bdb56373e2e9...
Subject Key Identifier	d58971b8a4bf5d7475d8f9eb...
Key Usage	Digital Signature, Key Encipher...
Thumbprint	b9844def377382027f2f646516803bef714f22ee

b9844def377382027f2f646516803bef714f22ee

Actions
Certificates
More Actions
WEC-Marcin
More Actions

Administrator: Console1 - [Console Root\Certificates (Local Computer)\Personal\Certificates]

File Action View Favorites Window Help

Console Root

- Certificates (Local Computer)
 - Personal
 - Certificates
 - Trusted Root Certification Authorities
 - Enterprise Trust
 - Intermediate Certification Authorities
 - Trusted Publishers
 - Untrusted Certificates
 - Third-Party Root Certificates
 - Trusted People
 - Client Authentication Issuance
 - Preview Build Roots
 - Test Roots
 - Other People
 - Certificate Enrollment Request
 - Smart Card Trusted Root
 - Trusted Packaged App Installation
 - Trusted Devices
 - Windows Live ID Token
 - WindowsServerUpdateS

Issued To	Issued By	Expiration Date	Intended Purposes	Friendly Name
Marcin-Lab-RootCA	Marcin-Lab-RootCA	12/08/2036	<All>	<None>
WEC-Marcin	Marcin-Lab-RootCA	12/08/2031	Server Authentication	<None>
WEF-Marcin	Marcin-Lab-RootCA	12/08/2031	Client Authentication	<None>

Actions

- Certificates
- More Actions
- WEF-Marcin
- More Actions

Certificate

General Details Certification Path

Show: <All>

Field	Value
Public key	RSA (2048 Bits)
Public key parameters	05 00
Enhanced Key Usage	Client Authentication (1.3.6.1.4.1.311.2.1.1)
Authority Key Identifier	KeyID=cab877b0db56373e2e9...
Subject Key Identifier	9710e5c725f895eaa293399b4...
Key Usage	Digital Signature, Key Encipherment
Thumbprint	ebcf3d97349de1e321a32056c3dc65c8218d15c6

ebcf 3d97349de1e321a32056c3dc65c8218d15c6